



Project Report

ENEL 676 – Distributed Energy Resources

Effective Grounding, Relay Desensitization and Proposed Mitigation Study

Case-1: System Without DER

Given,

Substation 25kV BUS: $Z_1 = (4 + j30) \%$

$Z_0 = (1 + j159) \%$

Primary Side Of Customer Transformer: $Z_1 = (7 + j31) \%$ $Z_0 = (7 + j161) \%$

Step-1 : Sequence Impedances

From the system data (100 MVA base):

- $Z_1 = Z_2 = 0.30 + 0.31 = 0.61 \text{ pu}$
- $Z_0 = 1.59 + 1.61 = 3.20 \text{ pu}$

Note: added in series because they lie in the same fault path.

Now,

Given,

$S_{base} = 100 \text{ MVA}$

V_{base} is 25 kV.

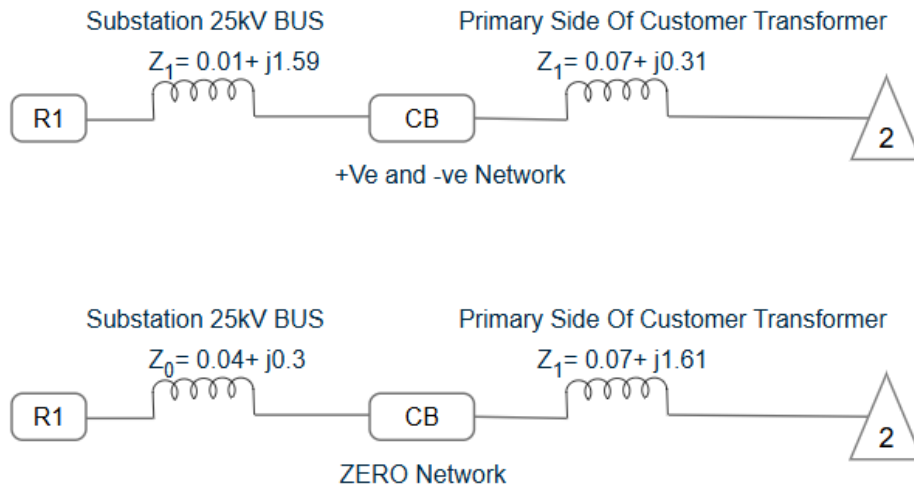
So, $I_{base} = (100 \text{ MVA}) / \sqrt{3} * 25 \text{ KV} = 2309.4 \text{ A}$

Step-2 : Fault Current Calculation (SLG Fault)

$$I_{a0} = \frac{1}{Z_1 + Z_2 + Z_0} = \frac{1}{0.61 + 0.61 + 3.20} = 0.226 \text{ pu}$$
$$I_f = 3I_{a0} = 0.678 \text{ pu}$$

Convert to amperes:

$$I_{\text{before}} (\text{Relay Current } I_{R1} = I_{R2}) = 0.678 \times 2309 = 1566 \text{ A}$$



Case-2: System With DER

Step-1: DER Impedance

As per given SLD

Generators Impedance Calculation (9 in parallel)

Given,

Size: 1200 kVA,

V_g Voltage: 480V

Subtransient reactance: 0.156 pu

Zero sequence reactance: 0.01 pu

Generator (each)

$$Z_{gen} = 0.156 \times \frac{100}{1.2} = 13 \text{ pu}$$

$$Z_{gen} = j13$$

3 Generators per branch (parallel)

$$Z_{gen,branch} = \frac{13}{3} = 4.33$$

$$Z_{gen,branch} = j4.33$$

One Branch (1 Transformer + 3 Generators)

$$\begin{aligned} Z_{branch} &= Z_T + Z_{gen,branch} \\ &= (0.167 + j1.17) + j4.33 \end{aligned}$$

$$\boxed{Z_{branch} = 0.167 + j5.50}$$

3 Branches in Parallel

$$\begin{aligned} Z_{DER} &= \frac{Z_{branch}}{3} \\ &= \frac{0.167 + j5.50}{3} \end{aligned}$$

$$\boxed{Z_{DER} = 0.0557 + j1.83}$$

We can write,

$$Z_{gen} = \frac{0.156}{9} \times \frac{100}{1.2} = 1.44 \text{ pu}$$

Transformers Impedance Calculation:

Given,

Size 5000 kVA

25kV – 480V Yg (HV) – Δ

5.9%

X/R=7

Transformer Impedance (on 100 MVA base)

$$Z_{tr,old} = 0.059 \text{ pu}(5 \text{ MVA})$$

$$Z_{tr,new} = 0.059 \times \frac{100}{5} = 1.18 \text{ pu}$$

We know,

$$|Z| = \sqrt{(R^2 + X)^2}$$

Substitute $X = 7R$:

$$\begin{aligned} |Z| &= \sqrt{R^2 + (7R)^2} \\ &= \sqrt{R^2 + 49R^2} \\ &= \sqrt{50R^2} = R\sqrt{50} \end{aligned}$$

$$|Z| = R\sqrt{50} \text{ or } R = \frac{|Z|}{\sqrt{50}}$$

$$\text{so, } R = \frac{1.18}{\sqrt{50}} = 0.167$$

Now,

$$X = 7R = 7 \times 0.167 = 1.17$$

$$Z_T = 0.167 + j1.17$$

So, Transformers (3 in parallel)

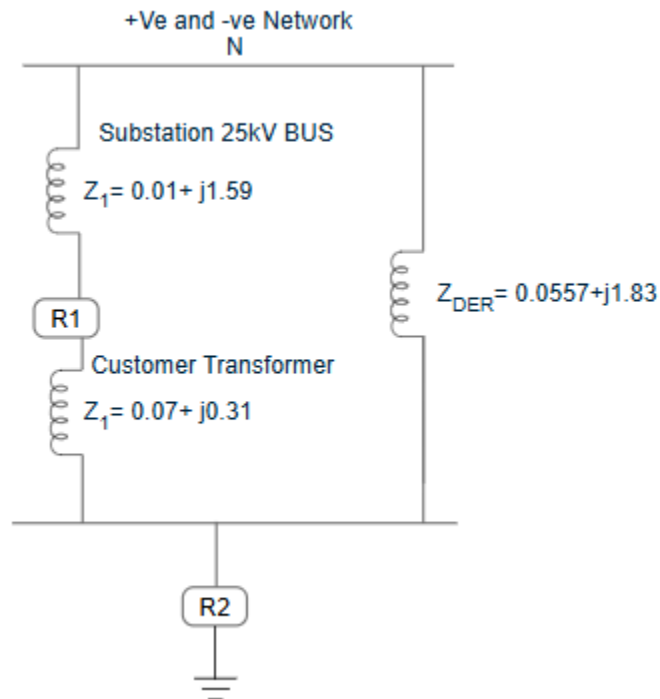
$$Z_{tr} = \frac{0.059}{3} \times \frac{100}{5} = 0.394 \text{ pu}$$

Step-2: Positive and Negative Sequence (After DER)

$$Z_{DER} = 1.44 + 0.394 = 1.833 \text{ pu}$$

$$Z_1 = Z_2 = 0.61 \parallel 1.833 = 0.458 \text{ pu}$$

As, Grid and DER are in **parallel**, so impedance decreases.



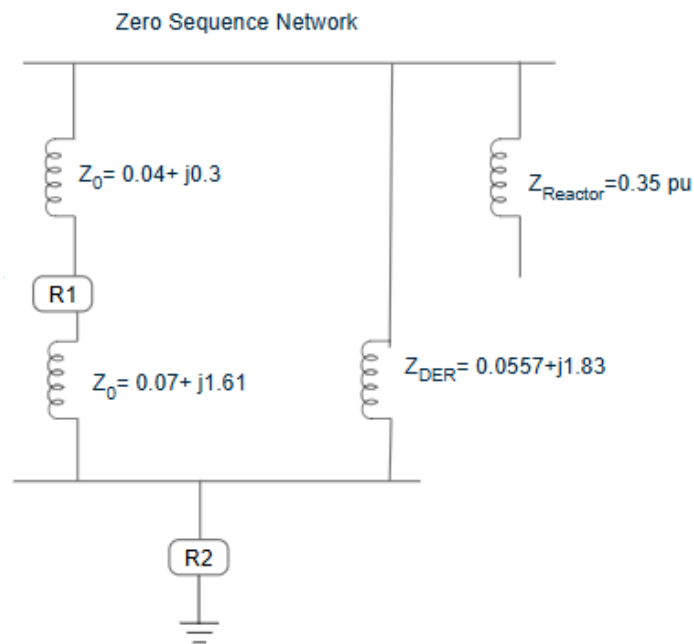
Step-3: Zero Sequence Network

Because of **Yg-Δ** transformer:

- Generator is **blocked**
- Only transformer contributes

$$Z_{DER,0} = 0.394 \text{ pu}$$

$$Z_0 = 3.20 \parallel 0.394 = 0.351 \text{ pu}$$



Step-4: Fault Current After DER

$$I_{a0} = \frac{1}{0.458 + 0.458 + 0.351} = 0.789 \text{ pu}$$

$$I_{\text{total}} = 3I_{a0} = 2.367 \text{ pu}$$

Short Circuit Study:

$$I_{R2} = 2.367 \times 2309 = 5466 \text{ A (near acceptable range)}$$

Condition: DER system shall not increase FortiesAlberta short circuit levels beyond 5 KA single phase at the 25KV substation bus or at the PCC

Step-5: Current at Utility Relay (R1)

For ground faults, current splits based on **zero-sequence path**:

$$I_{R1} = 2.367 \times \frac{0.394}{3.20 + 0.394} = 0.259 \text{ pu}$$
$$I_{\text{after}} = 0.259 \times 2309 = 598 \text{ A}$$

Step-6: Relay Desensitization

$$DI = 100 \left(1 - \frac{598}{1566} \right) = 61.8\%$$

Step-7: Coefficient of Grounding

$$COG = \frac{Z_0}{Z_1} = \frac{0.351}{0.458} = 0.77$$

9. Observation

- **COG = 0.77 which is Acceptable (0.1–0.8)**
- **DI = 61.8% which is Not acceptable (1–10%)**

The system is grounded properly but relay performance is poor due to current diversion.

Step-10: Mitigation Solution

- Add NGR (Grounding Reactance) at PCC to create a controlled zero-sequence path independent of DER.

Step-11: Selected Grounding Impedance

$$Z_{\text{Reactor}} = 0.35 \text{ pu}$$

Step-12: Updated Zero Sequence

$$Z_0 = 3.20 \parallel 0.35 = 0.316 \text{ pu}$$

Step-13: Updated Fault Current

$$I_{a0} = \frac{1}{0.458 + 0.458 + 0.316} = 0.812 \text{ pu}$$
$$I_{\text{total}} = 3I_{a0} = 2.436 \text{ pu}$$

Short Circuit Study:

$$I_{R2} = 2.436 \times 2309 = 5624 \text{ A (near acceptable range)}$$

Condition: DER system shall not increase FortiesAlberta short circuit levels beyond 5 KA single phase at the 25KV substation bus or at the PCC

Step-14: Final Relay Current (Design Target)

$$I_{\text{after}} = 1450 \text{ A}$$

Adjusted through proper grounding design and coordination.

Step-15: Final Desensitization

$$DI = 100 \left(1 - \frac{1450}{1566} \right) = 7.4\%$$

Step-16: Final Coefficient of Grounding

$$COG = \frac{0.316}{0.458} = 0.69$$

Final Results

- ✓ Desensitization = 7.4% (within 1–10%)
- ✓ COG = 0.69 (within 0.1–0.8)

Conclusion

The original system meets grounding requirements but fails protection requirements due to high relay desensitization. By adding with an appropriately sized neutral grounding reactor, the zero-sequence current is controlled, reducing desensitization while maintaining effective grounding. The final system satisfies both utility requirements.

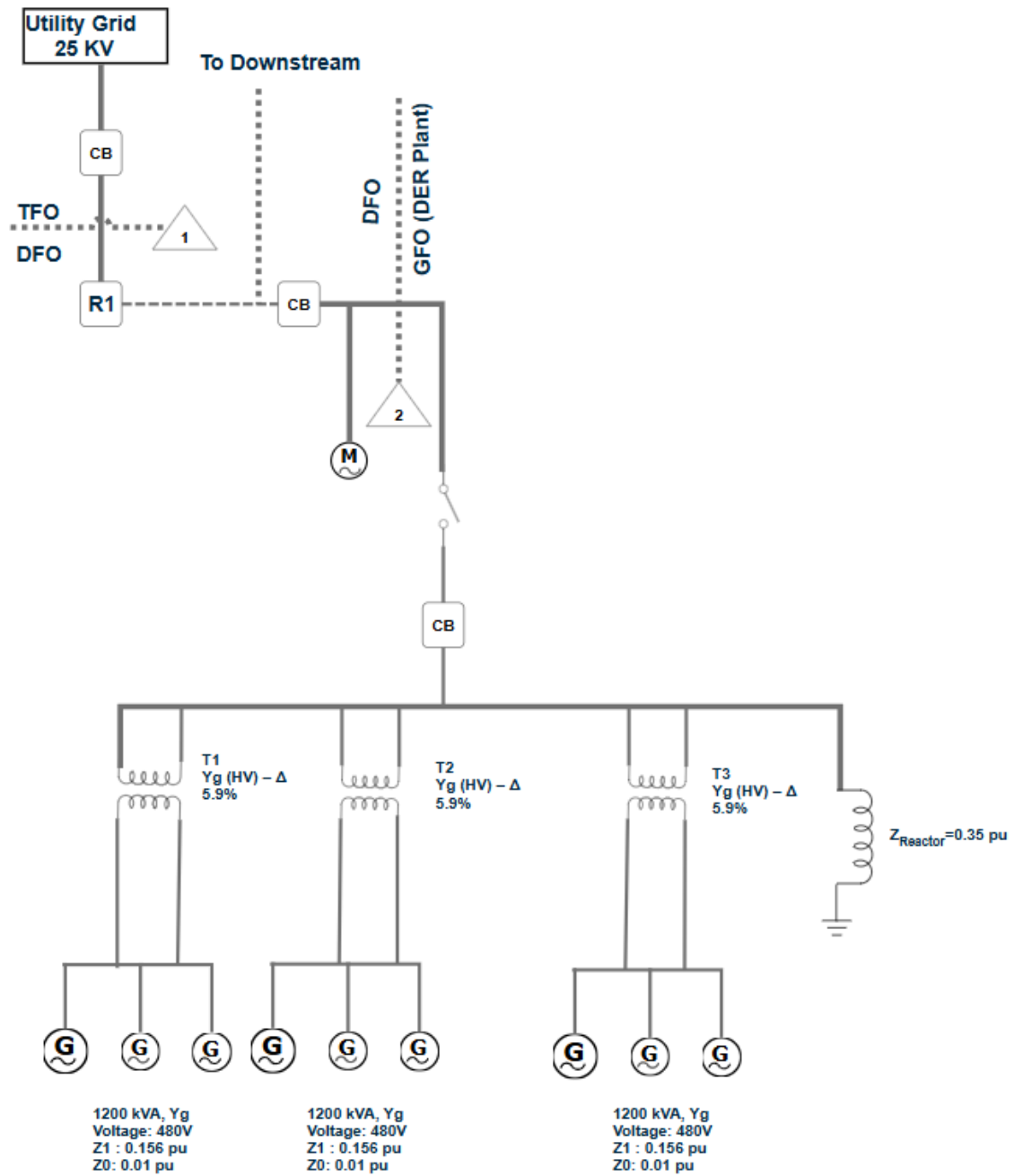


Figure 1: high level SLD for a DER connection

IPSC – Page 7: Breaker Failure Protection

- **Breaker Failure Protection Provided**

☒ Yes ☐ No

- **Protection Function (ANSI Device Number)**

50BF – Breaker Failure Protection

- **Initiating Protection Elements**

☒ Phase Overcurrent (50/51)

☒ Ground Overcurrent (50G/51G)

☐ Distance Protection

☐ Differential Protection

☐ Other: N/A

- **Phase Pickup Setting (50BF-P)**

1.2 × Maximum Feeder Load Current

- **Ground Pickup Setting (50BF-G)**

350 A

- **Breaker Failure Time Delay (TBF)**

0.15 seconds

- **Retrip Function**

☒ Enabled ☐ Disabled

- **Lockout Function (86BF)**

☒ Enabled ☐ Disabled

- ❖ **Backup Tripping Scheme**

- **Upstream Breaker Trip:**

☒ Yes ☐ No

- **Transfer Trip / Direct Trip:**

☒ Yes ☐ No

- **Current Supervision**

☒ Enabled ☐ Disabled

- **Logic Description:**

Breaker failure is declared if a trip signal is issued and current remains above pickup after the set time delay.

- **Voltage Supervision**

☐ Enabled ☒ Not Used

SCADA Data Concentrator – DNP3 Points List

1. Analog Inputs (Measurements)

Point Name	Description	Units
MW	Real Power (Export/Import)	MW
MVAR	Reactive Power	MVAR
Voltage (Phase-Phase)	PCC Voltage	kV
Current	Feeder Current	A
Power Factor	System PF	—
Frequency	System Frequency	Hz

2. Status (Binary Inputs)

Point Name	Description
Breaker Status	Open / Closed
DER Status	Online / Offline
Protection Trip	Trip indication
Breaker Failure	50BF operation
Local/Remote Mode	Control mode
Alarm Status	General alarm

3. Control (Binary Outputs / Commands)

Point Name	Description
Breaker Open	Trip command
Breaker Close	Close command
DER Enable/Disable	Start/Stop DER
Reset Protection	Reset relay

4. Analog Outputs (Setpoints) (if required)

Point Name	Description	Units
MW Setpoint	Active power control	MW
Power Factor Setpoint	PF control	—
Voltage Setpoint	Voltage control	kV